

IQR formula Explanation

Outlier Formula Explanation

Outliers are extreme or unusual values that differ significantly from the rest of the dataset. The **Interquartile Range (IQR)** method is commonly used to detect outliers.

Formulas

Lesser Outlier	Greater Outlier	IQR Formula
$Q1 - 1.5 * IQR$	$Q3 + 1.5 * IQR$	$IQR = Q3 - Q1$

Where:

- **Q1** = 25th percentile
- **Q3** = 75th percentile
- **IQR** = Interquartile Range

Why is 1.5 Used?

The value **1.5** is a standard statistical multiplier used to create boundaries for detecting abnormal data points.

- Values below the lower limit are called **lower outliers**
- Values above the upper limit are called **upper outliers**
- It helps identify unusual, incorrect, or rare observations in the dataset

Example Suppose:

- $Q1 = 10$
- $Q3 = 20$

Then:	Lower Limit:	Upper Limit:
$IQR = 20 - 10 = 10$	$10 - 1.5(10) = -5$	$20 + 1.5(10) = 35$

So:

- Values below **-5** are lower outliers
- Values above **35** are upper outliers

Conclusion

The **$1.5 \times IQR$** rule is widely used in statistics and data science because it effectively identifies extreme values while keeping normal data within the acceptable range.